Compounds of Transitional elements

101. Which of the following compounds volatilises on heating

(a) $MgCl_2$ (b) $HgCl_2$

(c) $CaCl_2$ (d) $FeCl_3$

102. Which of the following statement is not true

(a) Colourless compounds of transition elements are paramagnetic

(b) Coloured compounds of transition elements are paramagnetic

(c) Colourless compounds of transition elements are diamagnetic

(d) Transition elements form the complex compounds

103. Calamine is a mineral, which is

(a) $ZnCO_3$ (b) ZnS

(c) $ZnSO_4$ (d) ZnO

104. Super conductors are derived from compounds of

(a) *p*-block elements

(b) Lanthanides

(c) Actinides

(d) Transition elements

105. Manganese achieves its highest oxidation state in its compound

(a) MnO_3 (b) Mn_3O_4

(c) $KMnO_4$ (d) K_2MnO_4

106. Which can be reduced to the metal by heating it in a stream of hydrogen

(a) Copper (II) oxide

(b) Magnesium oxide

(c) Aluminium oxide

(d) Calcium oxide

107. Which of the following is coloured

(a) $ScCl_3$ (b) TiO_2

(c) $MnSO_4$ (d) $ZnSO_4$

108. **Chrome green** is

(a) Chromium sulphate

(b) Chromium chloride

(c) Chromium nitrate

(d) Chromium oxide

109. The colour of $(NH_4)_2SO_4Fe(SO_4)_3.24H_2O$ is

(a) White (b) Green

(c) Violet (d) Blue

110. Correct formula of potassium ferricyanide is

(a) $K_4Fe(CN)_6$

(b) $K_3Fe(CN)_6$

(c) $K_3[Fe(CN)_3]$

(d) $K_3[Fe(CN)_4]$

111. The form of iron having the highest carbon content is

(a) Cast iron

(b) Wrought iron

(c) Strain steel

(d) Mild steel

112. Aqueous solution of ferric chloride is

(a) Acidic

(b) Basic

(c) Neutral
(d) Amphoteric

113. In the reduction of dichromate by $Fe(II)$ the number of electrons involved per chromium atom is
(a) 2 (b) 3
(c) 4 (d) 1

114. A group of acidic oxide is
(a) CrO_3, Mn_2O_7 (b) ZnO, Al_2O_3
(c) CaO, ZnO
(d) Na_2O, Al_2O_3

115. Silver nitrate is mainly used
(a) In photography
(b) In model formation
(c) As reducing agent
(d) As dehydrating agent

116. The correct order of magnetic moments (spin only values in B.M.) among is
(a) $[Fe(CN)_6]^{4-} > [MnCl_4]^{2-} > [CoCl_4]^{2-}$
(b) $[MnCl_4]^{2-} > [Fe(CN)_6]^{4-} > [CoCl_4]^{2-}$
(c) $[MnCl_4]^{2-} > [CoCl_4]^{2-} > [Fe(CN)_6]^{4-}$
(d) $[Fe(CN)_6]^{4-} > [CoCl_4]^{2-} > [MnCl_4]^{2-}$
(Atomic nos. $Mn = 25, Fe = 26, Co = 27$)

117. Hybridization of $[Ni(CO)_4]$ is
(a) sp^3 (b) d^2sp^3
(c) sp^3d (d) sp^2

118. What is the oxidation number of iron in the compound $[Fe(H_2O)_5(NO)]SO_4$
(a) $+2$ (b) $+3$
(c) ±1 (d) ±4

119. Which of the following metal gives hydrogen gas, when heated with hot concentrated alkali
(a) *Cu* (b) *Ag*
(c) *Zn* (d) *Ni*

120. When ferric oxide reacts with *NaOH*, the product formed is
(a) *NaF* (b) $FeCl_3$
(c) $Fe(OH)_3$ (d) $NaFeO_2$